Dear Prof. Wu and Selection Committee,

I am writing to express my strong interest in the PhD position in Adipose Tissue Inflammation and Metabolic Health in your laboratory at the Department of Biomedical Sciences, University of Lausanne. This project deeply resonates with my motivation to unravel the molecular mechanisms of metabolic diseases and to contribute to translational research that bridges fundamental biology with clinical impact.

I hold a BSc in Biology and an MSc in Medical Biology, where I gained a solid foundation in molecular and cellular physiology, neuroscience, and biomedicine. Through these programs and subsequent research positions, I acquired extensive experience in molecular biology and biochemistry, including PCR/qPCR, cloning, western blotting, ELISA, immunohistochemistry, microscopy, RNA/DNA extraction, and cell and tissue culture.

During my BSc and Junior Research Assistant work at MSU, I investigated vascular tone regulation using *ex vivo* artery models in rats, combining wire myography, molecular techniques, and statistical analysis. I also contributed to a preeclampsia project and an artery cultivation study, which further developed my expertise in handling delicate tissue samples, designing experiments, interpreting functional data, and reporting results. Later, during my MSc internship at the Donders Institute, I employed two-photon calcium imaging to study neuronal activity *in vivo* in mice, complemented by computational analysis in Python and DeepLabCut. These experiences reinforced my ability to integrate experimental and systems-level approaches to generate biological insight.

Beyond academic research, I broadened my translational perspective by working at a Contract Research Organization, where I contributed to the design and planning of clinical trials, produced extensive literature reviews, and ensured compliance with regulatory frameworks. This exposure gave me a strong appreciation of how mechanistic findings are translated into therapeutic strategies.

What excites me most about this PhD is the opportunity to combine wet-lab experimentation with systems biology approaches to better understand adipose tissue inflammation and its metabolic consequences. I am especially motivated by the integration of animal models, human cohorts, advanced proteomics, and multi-omics analyses to identify molecular pathways that may ultimately inform new therapeutic strategies. I am eager to contribute to the scientific experiments, data analysis, and manuscript writing, as I see this position as a vital step toward my career goal of becoming a scientist specialized in molecular metabolism.

I am fluent in English, highly motivated, and eager to expand my expertise to include proteomics and large-scale data integration. I also hold a FELASA-accredited certificate (Art. 9), which allows me to work with animal models in the Netherlands, and I am willing to complete any additional training required to confirm my expertise.

Thank you for considering my application. I would be honoured to contribute to your team's innovative research on adipose tissue and metabolic health, and I look forward to the opportunity to discuss how my background and skills align with your group's goals.

Warm regards,
Varvara Lazarenko